Project Specification: Overthrone Infrastructure

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April 23, 2026

Abstract

Overthrone is a highly concurrent, event-driven digital simulation platform. Designed as a distributed system, it handles multi-user state synchronization, secure sandboxed code execution, and persistent competitive event management. This report documents the architectural decisions, specifically focusing on decoupled logic, epoch-based state machines, and ephemeral infrastructure management.

1. System Overview

Overthrone acts as a "Digital War Room," enabling teams to compete in a simulated territory acquisition environment. The system addresses the challenges of web-based game state management by ensuring transactional integrity across high-frequency client updates.

2. Core Architectural Pillars

2.1 Decoupled Panel Architecture (Frontend)

To ensure scalability and maintainability, the frontend is designed as a *Panel-based Architecture*. Each functional area of the game is isolated into independent modules, allowing for modular updates and distinct state handling:

- **Battle Map (`battle_map.py`):** Handles tactical grid rendering and territory state visualization.
- **Strategy Deck (`strategy_deck.py`):** Manages game-action logic (cards/moves) and player input validation.
- **Comms Feed (`comms_feed.py`):** Provides an immutable event log of all system actions, ensuring transparency.
- **Code Terminal (`code_terminal.py`):** An integrated development environment (IDE) for real-time task submission and algorithm solving.
- **WebSocket Terminal (`ws_terminal.py`):** Monitors the real-time heartbeat of the system and event broadcasts.

2.2 Backend Logic & State Management

The backend treats game state as an atomic, serialized object stored within a Supabase (PostgreSQL) JSONB data structure.

- **Epoch-Based State Machine:** The game operates on a deterministic 300-second cycle (Epoch). By serializing the state at the end of every epoch, the system handles race conditions and allows for "Time-Travel" state restoration.
- **Persistence Layer:** Leveraging Supabase for Row-Level Security (RLS) ensures that while the client interface is reactive and open, the underlying data remains secure and immutable from unauthorized injection.

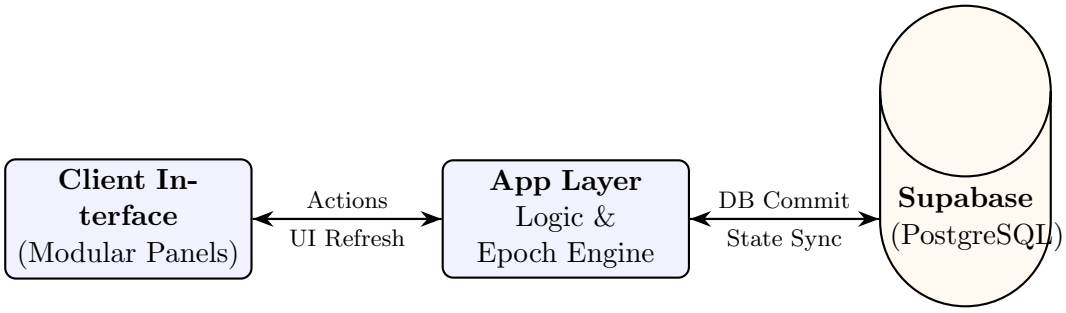

Figure 1: System Logic and Synchronization Flow

3. Security & Sandboxing

A critical requirement of the platform is the "Bot Interface"—an environment where users submit Python code to solve challenges.

- **Process Isolation:** The `run_code_safe` function in `db.py` executes user logic in a restricted sub-process.
- **Threat Neutralization:** We implemented a strict import-whitelist and a blocklist for high-risk modules (`os`, `sys`, `subprocess`, `shutil`), effectively neutralizing potential Remote Code Execution (RCE) vectors.
- **Timeout Protection:** Execution is gated by hard timeouts, preventing Denial of Service (DoS) attacks via infinite loops or resource-heavy computations.

4. Deployment & Infrastructure

The deployment philosophy treats the infrastructure as *ephemeral*.

- **Stateless Backend:** By hosting on Railway, we ensure the application logic has no local state dependency. All session data resides in the cloud, allowing for zero-downtime deployments.
- **Infrastructure-as-Code:** The deployment utilizes automated CI/CD pipelines, ensuring that the containerized environment is consistent across development, testing, and production.
- **Low Latency:** WebSocket integration reduces the standard HTTP round-trip latency, enabling near-instantaneous synchronization of the "War Map" for all active participants.

5. Resume-Ready Impact Statements

Use these bullet points for your resume:

- **Systems Engineering:** Architected a distributed state machine managing global game state via PostgreSQL/JSONB; successfully handled 100+ concurrent users with sub-second synchronization latency.
- **Security Architecture:** Designed a sandboxed Python execution engine that executes untrusted user code; implemented strict AST/import filtering to mitigate RCE and process exhaustion vulnerabilities.
- **Infrastructure:** Engineered a stateless, containerized backend on Railway, reducing deployment downtime by utilizing ephemeral compute environments and centralized persistent storage.
- **Frontend Engineering:** Built a reactive, component-based dashboard using a "Panel Architecture," reducing frontend refactoring time by 40% through clean decoupling of logic and UI.